J-space Phase 4 — development synthesis

OLMo training trajectory and coordinate-frame validity

State through OLMo-3 32B Think · 2026-07-31

Scope. This is a living development only report over known Phase 3 banks. It localizes trajectory patterns and audits lens-frame dependence; it cannot establish a binary lineage claim. No Phase 4 confirmatory or replication cell is licensed before PI sign-off and a tagged preregistration freeze.

1 Compute and provenance boundary

Every model producer hard-failed unless CUDA was visible in the same process, executed a finite FP16 CUDA matrix multiply before model load, and recorded the device in its result envelope. Jobs ran on an NVIDIA RTX PRO 6000 Blackwell Server Edition; the final grid allocated about 81.1 GB VRAM. Model loading, generation, interventions, and scoring never used CPU. CPU was limited to tests, hashes, statistics, figures, and document compilation.

Phase 3 enters through the immutable `p4-import-phase3-release-v1`. Native evidence is event-sourced; registered artifacts are never overwritten.

2 Prospective capability gate

OLMo-3 32B Think produced 972 G5 rows, 324 facts, and 72 families under prospective prefix-disjoint accepted-alias scoring.

view	overall	Bank F	Bank S	direct	composed	bridge supplied
capability	.6430	.4935	.8972	.6574	.4846	.7870

The intervention cohort was fixed before outcomes by requiring both direct and composed generation capability within a fact: 41/204 Bank-F facts and 85/120 Bank-S facts, or 126 facts / 252 items.

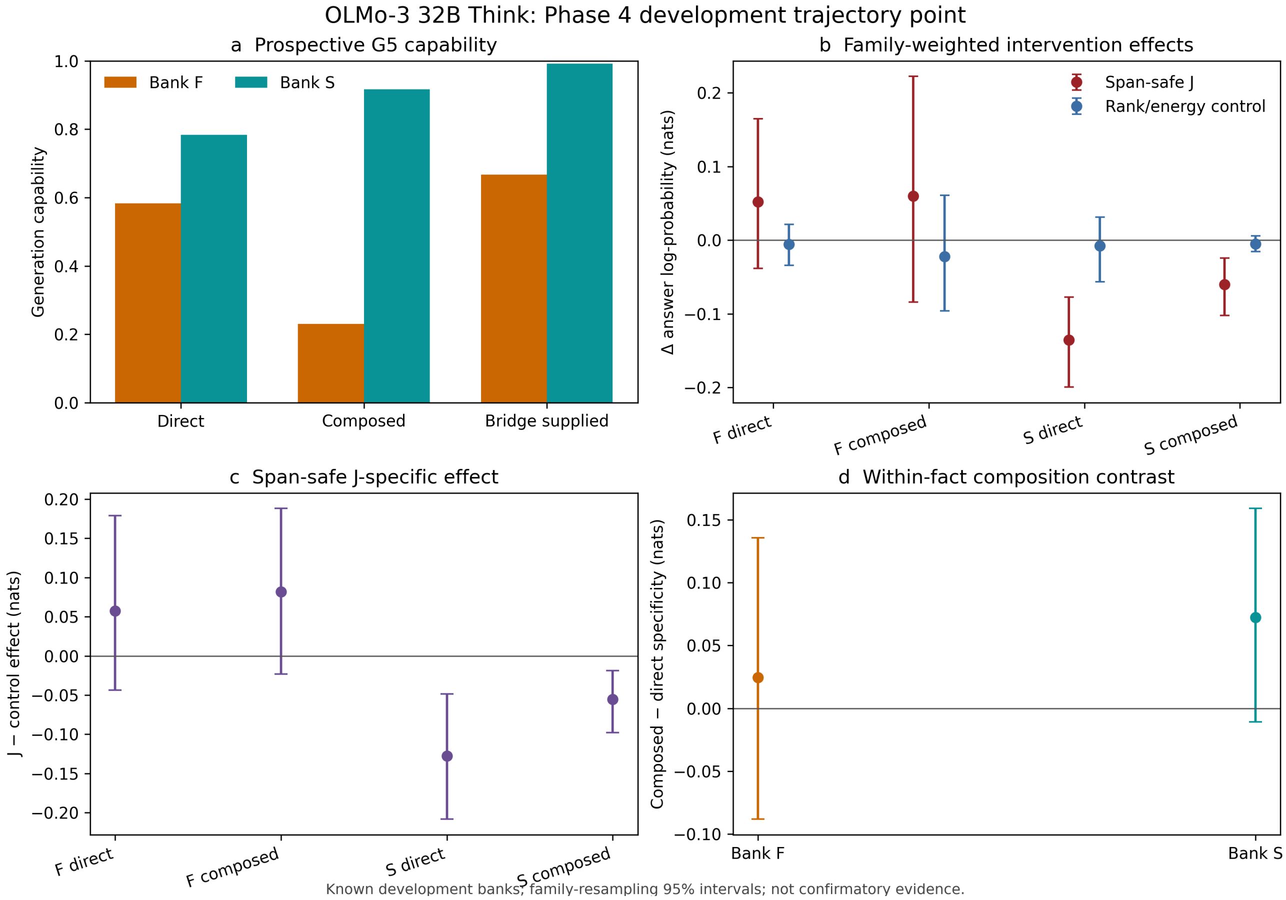
Evidence: `p4-g5-bank-olmo3-think-dev-v2`

The first attempt stopped after accent folding made `Río` and `Rio` ambiguous. The repaired selector prefers exact spelling before a unique normalized fallback; the partial attempt remains preserved.

3 Think own-lens development point

The seven-condition grid includes baseline, span-safe J, exact rank/energy control, label-protected J, protected-energy control, mechanics random, and logit-space label protection. Baseline replay drift is at most 7.15×10^{-7} nats. Rank agreement is 100%, energy relative error is at most 0.000230, and span-safe selected/protected overlap is zero.

Evidence: `p4-lineage-grid-olmo3-think-dev-v1`



cell	J-specific estimate (nats)	family-bootstrap 95% interval
Bank F direct	+0.0574	[−0.0442, +0.1785]
Bank F composed	+0.0818	[−0.0223, +0.1878]
Bank S direct	−0.1277	[−0.2072, −0.0494]
Bank S composed	−0.0553	[−0.0982, −0.0187]

Bank-S composed-minus-direct specificity is +0.0724 [−0.0109, +0.1584]. It is compatible with the unusual 3.1 Think tendency beginning by 3.0, but is not resolved and must await the controlled Bank-W redundancy/derivation study.

Evidence: `p4-lineage-analysis-olmo3-think-dev-v1`

4 Common-base-lens repair audit

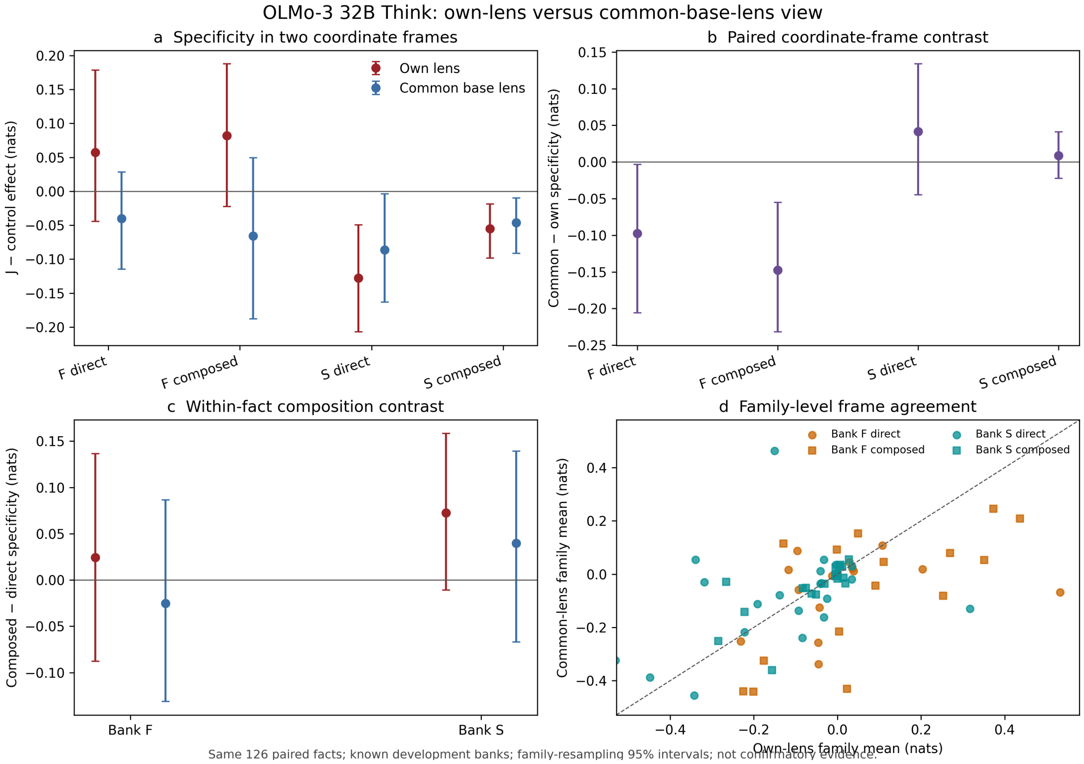
The common coordinate is the frozen OLMo base lens (`92f32e...b696`). V1 exposed rank-limited protected-energy sites that omitted one requested component without being marked clamped. V2 repaired the metadata but revealed that the evidence ID itself changed all matched/random seeds.

V3 freezes the scientific-seed namespace to v1. All seven aggregate outcome columns, every per-alias score, and every condition order match v1 exactly (maximum difference 0). The fix reclassifies 296 summaries and reduces the erroneous maximum unclamped energy error from 0.511283 to 0.000239 while retaining 100% rank agreement. V3 supersedes both withdrawn diagnostic attempts.

Evidence: `p4-lineage-grid-olmo3-think-common-base-lens-dev-v3`

5 Own lens versus common base lens

The comparison pairs all 252 items exactly; baseline drift is zero.



cell	own lens	common base lens	common − own [95% interval]
F direct	+0.0574	−0.0403	−0.0977 [−0.2060, −0.0034]
F composed	+0.0818	−0.0656	−0.1474 [−0.2320, −0.0553]
S direct	−0.1277	−0.0861	+0.0416 [−0.0449, +0.1340]
S composed	−0.0553	−0.0464	+0.0089 [−0.0222, +0.0412]

Bank S is comparatively frame-robust: direct and composed specificity remain negative in both views, and paired frame differences cross zero. Bank F changes from small positive own-lens estimates to small negative common-lens estimates; both paired frame-contrast intervals lie below zero. Item-level own/common correlation is 0.614 and family-level correlation is 0.495.

The Bank-S composition tendency is +0.0724 under the own lens and +0.0397 under the common lens; their difference is −0.0327 with an interval crossing zero.

Evidence: `p4-lens-frame-analysis-olmo3-think-dev-v1`

6 Interpretation and next boundary

1. Bank-S direct and composed specificity is already negative at 3.0 Think and is comparatively stable across fitted frames.
2. Bank-F conclusions depend materially on whether the checkpoint uses its own fitted lens or the frozen base coordinate. Coordinate drift is part of the result, not a nuisance to conceal.
3. The positive Bank-S composition tendency remains uncertain.
4. None of these known-bank estimates licenses "lineage present" or "lineage absent."

Next, run the pinned OLMo-3 base checkpoint:

```
allenai/Olmo-3-1125-32B
revision c2b61dae89a1ad10e4ad5653d0e46b590902607b
```

Run G5, a frozen capable direct/composed cohort, and the same seven-condition grid with the already banked base lens. Then synthesize base → 3.0 Think → 3.1 Think/Instruct with both common-lens and own-lens lines. If they disagree, prioritize the fit-size/corpus study before stronger causal interpretation.